

KENIL JITENDRABHAI KALATHIYA

M.Sc. Automotive Software Engineering

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PROFESSIONAL SUMMARY

Master's student in **Automotive Software Engineering** focusing on **ADAS**, simulation, driving systems, testing and **functional safety (ISO 26262)**. Practical experience with **ROS2**, Linux, Python, and distributed software systems. Goal: Entry into automotive-related software development.

TECHNICAL SKILLS

Automotive & Embedded Software: ROS2, Linux, ADAS Basic, CAN & Ethernet

Model-Based Design & Simulation: MATLAB, Simulink

Functional Safety: ISO 26262 (Basic)

Computer Vision & AI: OpenCV, YOLOv8, NumPy, TensorFlow / PyTorch

Web Development: React.js, Node.js, Vue.js

Database: MongoDB, SQL

Version Control & Development Tools: GitHub, GitLab, Jira

Programming Languages: Python, C, C++, JavaScript, TypeScript

EDUCATION

Apr 2024 – Current NOTE: Grade 2.3	M.Sc. Automotive Software Engineering at Technische Universität Chemnitz , Germany
Jun 2019 – Jul 2023 NOTE: Grade 1.5	B.Tech Information Technology at Gujarat Technological University , Ahmedabad, India

WORK EXPERIENCE

Apr 2026 – Present	Research Intern at Technische Universität Chemnitz - Designed a real-time structural health monitoring pipeline for UAV rotor arms using force-torque sensors in PX4 SITL and Gazebo Harmonic - Resolved non-functional strain gauge design by relocating sensors to arm root joints, achieving physically independent bending and torsion signals (correlation -0.17) - Built a ROS2 node computing bending, torsion and thrust at 100 Hz with filtering, logging to an 18-column self-labeled CSV dataset for ML training - Developed MAVSDK-scripted flight maneuvers and damage dataset plan (motor + mass degradation, 4 arms $\times$ 3 severities) for 1D-CNN/LSTM fault classification Tech Stack: Python, PX4 SITL, Gazebo Harmonic, ROS2, MAVSDK, QGroundControl, Force-Torque Sensing, Machine Learning, UAV Simulation
Oct 2022 – Apr 2023	Software Developer (Intern) at Bigscal Technologies Pvt. Ltd. — Surat, India - Developed modern frontend applications using React.js and JavaScript - Built reusable UI components - Integrated frontend modules with Flask-based REST APIs - Analyzed user requirements and optimized existing functions - Conducted unit, integration and acceptance tests Tech Stack: React.js, JavaScript, Flask, REST APIs

PROJECTS

Feb 2026 – Mar 2026	Intelligent Speed Adaptation (ISA) with Cruise Control — ADAS • Developed an Intelligent Speed Adaptation (ISA) system with state-based cruise control and PID-based speed control • Implemented zone-based speed limits for reliable ISA logic validation under simulator and hardware constraints • Integrated an RGB camera with real-time visualization and a preliminary perception pipeline for camera-based traffic sign recognition Tech Stack: Python, CARLA Simulator, OpenCV
Nov 2025	Vision-Based ADAS System — ADAS • Developed a camera-based ADAS system with lane detection and vehicle recognition • Implemented vehicle detection with YOLO and classical OpenCV-based image processing for robust lane detection • Computed Time-To-Collision (TTC) and implemented forward collision warning logic • Extended system with ROS2 nodes and basic camera–radar sensor fusion (simulation) Tech Stack: Python, OpenCV, YOLOv8, ROS2
Sep 2025 – Oct 2025	EV Energy Management & Real-Time Monitoring System • Developed a software-based energy management system for electric vehicles to simulate energy consumption and battery state of charge (SOC) • Implemented SOC-based Eco-Mode logic to optimize efficiency • Integrated the system into a ROS2 publisher–subscriber architecture for real-time transmission of energy signals (SOC, power, eco-mode status) • Used Linux (Ubuntu via WSL2) and the ROS2 build system Colcon to develop modular ROS2 nodes • Focus: EV fundamentals, vehicle energy management, ROS2 middleware Tech Stack: Python, ROS2, Linux, NumPy

STANDARDS AND PROCESSES

- **ISO 26262** — Functional Safety (Basics)
- V-Model / Systems Engineering
- Software Testing & Quality Practices

CERTIFICATIONS

Coursera Certificate: System Architecture [2022]

SOFT SKILLS

- Team leadership and project management
- Independent work and quick adaptability
- Problem-solving mindset
- Strong communication and teamwork

LANGUAGES

English (Fluent – C1)
Deutsch (B1)